Interpretable Machine Learning for Wine Quality Prediction

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***Abstract*—This paper addresses the inherent subjectivity in traditional wine quality assessment by developing an interpretable machine learning framework for objective quality prediction. Using the UCI Wine Quality dataset containing 6,497 samples with 11 physicochemical properties, we implement and optimize a Random Forest classifier that balances accuracy with practical interpretability. Our baseline model achieves 83.69% accuracy but exhibits limited performance on minority classes. Through Synthetic Minority Oversampling Technique (SMOTE) and hyperparameter optimization, our enhanced model (81.3% accuracy) demonstrates substantially improved detection of underrepresented quality categories, increasing macro-average F1-score from 0.52 to 0.55. Feature importance analysis identifies alcohol content, volatile acidity, and free sulfur dioxide as the most influential quality predictors—findings consistent with oenological domain knowledge. We emphasize visual interpretability through informative visualizations and establish a reproducible pipeline suitable for real-world deployment. This framework bridges data science and practical oenology, offering an objective complement to sensory evaluation that benefits producers, distributors, and consumers in the $400 billion wine industry.**

***Index Terms*—wine quality prediction, machine learning, random forest, class imbalance, SMOTE, interpretable machine learning, enology, food quality assessment**

I. Introduction

The global wine industry, valued at over $400 billion, is undergoing rapid growth and diversification. As new markets emerge and consumer expectations evolve, there is a rising demand for objective, scalable, and consistent quality control mechanisms. Traditionally, wine quality assessment relies heavily on human sensory evaluation, involving professional tasters and sommeliers. While this method is deeply rooted in wine culture, it is inherently subjective, often inconsistent, and can vary widely based on the evaluators experience, fatigue, or personal preferences. This subjectivity poses practical challenges for producers, distributors, and consumers alike. For producers, it affects pricing strategies, product positioning, and brand trust. For consumers, it introduces uncertainty in purchasing decisions and erodes confidence in quality claims. This study aims to address these limitations by introducing a data-driven framework for wine quality prediction that is interpretable, reproducible, and suitable for deployment at scale. Previous studies have applied classical machine learning methods to wine quality prediction but suffer from: (1) limited incorporation of domain knowledge, (2) insufficient emphasis on model interpretability, and (3) minimal attention to practical deployment [1]. This work addresses these gaps by applying supervised learning to predict wine quality from physicochemical properties, while emphasizing model transparency and reproducibility. The UCI Wine Quality dataset provides a foundation for analysis, it lacks key variables such as grape variety, vineyard location, soil characteristics, and climatic conditions that significantly influence wine quality in practice. Therefore, we are able to lay the groundwork for future integration of contextual factors (e.g., vineyard conditions) and propose a scalable, interpretable, and user-oriented modeling pipeline.

This study advances wine quality prediction through three key contributions:

- **Predictive Modeling:** We evaluate several machine learning algorithms on the UCI Wine Quality dataset, benchmarking performance using accuracy, precision, recall, and F1-score.
- **Visual and Practical Interpretability:** Beyond traditional metrics, we emphasize model interpretability through informative visualizations (e.g., heatmaps, SHAP-based feature rankings, and confusion matrices). We also consider deployment strategies such as web-based or mobile prediction interfaces.
- **Reproducibility and Scalability:** We implement a reproducible, well-documented ML pipeline using version control and modular code design. The framework is adaptable for future use in wine analytics or other agricultural domains.

In summary, this study contributes a robust and interpretable ML pipeline that balances predictive performance, usability, and deployment potential.

II. Related Work

Machine learning has gained traction in food and beverage quality assessment, including wine. Numerous studies have explored wine quality prediction using the UCI Wine Quality dataset, which contains physicochemical attributes (e.g., acidity, alcohol, pH, sulfates) and a quality score as the target. Prior work has applied algorithms such as Decision Trees, Random Forests, Support Vector Machines (SVMs), Logistic Regression, and K-Nearest Neighbors (KNN) for both classification and regression tasks. For example, Cortez et al. [2]

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applied SVMs and ensemble methods, reporting that Random Forests often outperform single classifiers when dealing with noise or imbalanced distributions. Other researchers, such as Jolliffe et al. [3], have employed dimensionality reduction techniques such as principal component analysis (PCA) or feature selection to improve model accuracy. Although deep learning methods have also been explored [1], their advantages are limited by the modest size and structured nature of the dataset, which favors traditional machine learning methods. While these approaches have achieved moderate success (typically 60–70% accuracy), Li et al. [1] often emphasize predictive performance over interpretability and seldom consider practical deployment. Furthermore, many treat the data as a black box, providing limited insight into which features truly drive quality from an oenological perspective. Recent research has shifted toward understanding model behavior, not just maximizing accuracy. Studies, including one by ZaZa et al. [4], consistently identify alcohol content, volatile acidity, and citric acid as key predictors of quality, aligning with sensory science literature on taste, mouthfeel, and consumer preference. However, few works translate these insights into domain-relevant visualizations or explanations accessible to non-technical stakeholders such as winemakers or quality control teams. Despite substantial progress, gaps remain: *Lack of practical tools* (most studies stop at evaluation without translating models into usable applications [5]), *Limited visualization strategies* (insights are rarely communicated through intuitive, domain-relevant visualizations [6]), and *Neglected contextual variables* (external factors such as vineyard location, soil, or climate are often excluded [7]). This study addresses these gaps by implementing interpretable ensemble-based models with meaningful visual outputs and designing a reproducible ML pipeline suitable for real-world deployment.

III. METHODOLOGY

A. Dataset and Preprocessing

This study utilized the UCI Wine Quality dataset [8], which contains physicochemical properties and quality ratings for both red and white wine varieties. The dataset is publicly available in CSV format and includes 11 input features (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulfates, and alcohol content) and a quality score target variable ranging from 3 to 9. A Random Forest classifier was selected for its robustness to overfitting, strong empirical performance on tabular data, and ability to provide interpretable measures of feature importance.

a) Data Cleaning and Imputation: Initial data quality assessment revealed 38 missing values distributed across several features: fixed acidity (10), volatile acidity (8), citric acid (3), residual sugar (2), chlorides (2), pH (9), and sulfates (4). We addressed these gaps using median imputation to preserve the statistical integrity of the dataset while minimizing the influence of outliers. This imputation strategy was particularly appropriate given the right-skewed distributions observed in several features, notably citric acid, volatile acidity, and fixed acidity, where extreme values could distort mean-based imputation approaches.

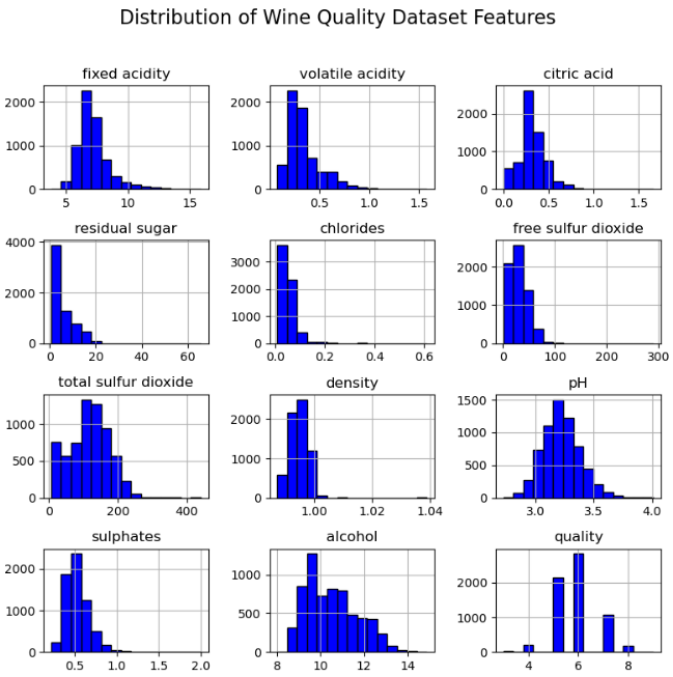


Fig. 1. Distribution of wine quality dataset features across physicochemical variables.

b) Exploratory Data Analysis: The dataset includes 11 physicochemical features, each summarized by count, mean, standard deviation, and percentiles. For instance, fixed acidity presented an average of approximately 7.22 with a standard deviation of 1.30. Features such as residual sugar ($\mu = 5.44$, $\sigma = 4.76$) and free sulfur dioxide ($\mu = 30.53$, $\sigma = 17.75$) exhibited particularly high variability. In contrast, pH ($\mu = 3.22$, $\sigma = 0.16$) remained relatively consistent across samples. The target variable, quality, had a mean of approximately 5.82 (ranging from 3 to 9) with a median (50th percentile) of 6, indicating a slight right skew in the ratings. Alcohol content averaged 10.5%. The wide ranges and high standard deviations observed in features like residual sugar, free sulfur dioxide, and total sulfur dioxide suggest potential heteroscedasticity or skewness. These statistical insights are crucial for guiding preprocessing steps such as normalization, transformation, and outlier management.

The distribution of features in the dataset, as shown in Fig. 1 reveals that most variables are right-skewed, indicating a concentration of lower values with occasional high outliers. Notably, variables such as *residual sugar*, *free sulfur dioxide*, and *total sulfur dioxide* exhibit long right tails, reflecting sporadically high concentrations. In contrast, features like *pH* and *quality* display more symmetric or discrete distributions. The *quality* variable, in particular, is discrete and concentrated around values 5, 6, and 7, which are the most frequently assigned ratings. These distributional characteristics highlight the importance of preprocessing steps such as normalization,

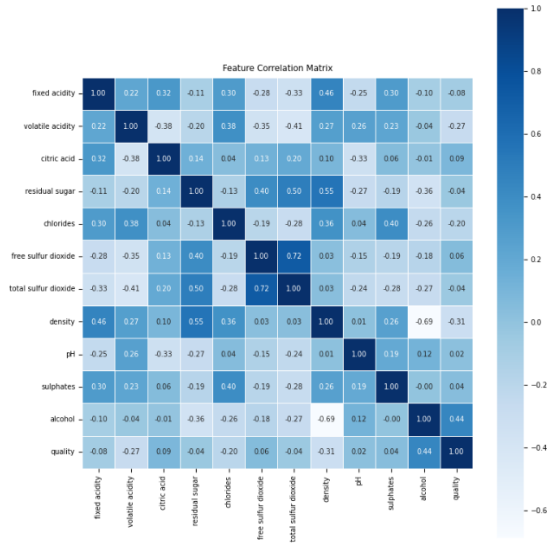


Fig. 2. Feature correlation matrix heatmap. Darker colors indicate stronger positive correlations, while the intensity reflects the magnitude of the correlation coefficient.

transformation, or binning before applying machine learning models. A frequency analysis of the quality ratings further confirms that quality score 5 appears most frequently, while all classes have at least 12 samples.

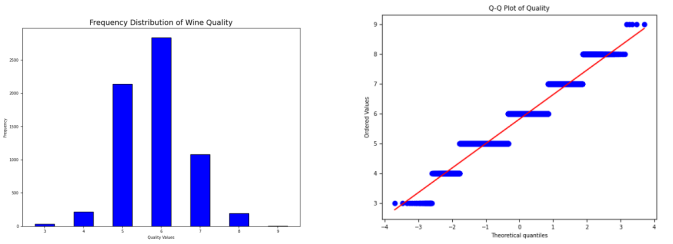
B. Correlation Analysis and Feature Selection

To identify potential multicollinearity among features, which could adversely affect model performance, a correlation analysis was conducted. A heatmap visualizing the feature correlation matrix was generated, as shown in Fig. 2.

The correlation matrix revealed a strong positive correlation between 'total sulfur dioxide' and 'free sulfur dioxide', with a correlation coefficient of 0.72. Such high correlation suggests redundancy between these two features. To mitigate potential multicollinearity and simplify the model, one of these features was selected for removal. The decision was based on the individual contribution of each feature to predicting wine quality, assessed using the coefficient of determination (R^2) from a simple linear regression against the 'quality' variable. 'Free sulfur dioxide' yielded an R^2 value of 0.0031, whereas 'total sulfur dioxide' had an R^2 value of 0.0017. Given its lower R^2 value, indicating a weaker individual relationship with wine quality, 'total sulfur dioxide' was removed from the dataset. Other variables generally exhibited lower inter-correlations, as indicated by lighter coloring and smaller correlation coefficients in the heatmap.

C. Model Selection and Training

a) Model Selection: To select an appropriate model for wine quality prediction, we first evaluated several assumptions based on the structure and distribution of the data. The target variable, *quality*, is multi-class and imbalanced, with ratings like 5 and 6 appearing most frequently, while others (e.g., 3 and 9) are underrepresented. This class imbalance made algorithms like Logistic Regression and Support Vector Classifiers less suitable due to their limitations in handling multi-class and imbalanced data effectively. Further diagnostic checks showed that the predictor variables do not follow a normal distribution (Fig. 3b) and contain several outliers. These observations ruled out models like Linear Discriminant Analysis and distance-based methods such as K-Nearest Neighbors and SVMs, which are sensitive to distributional assumptions and outliers. Given the data characteristics and our goal of optimizing classification accuracy, we selected the Random Forest algorithm. It is robust to non-normality, handles class imbalance better, and provides interpretable decision rules through its tree-based structure.



(a) Distribution of wine quality scores, showing skew toward mid-range ratings.

(b) Q-Q plot showing deviation from normality in wine quality scores.

Fig. 3. Exploratory visualizations of the wine quality target variable.

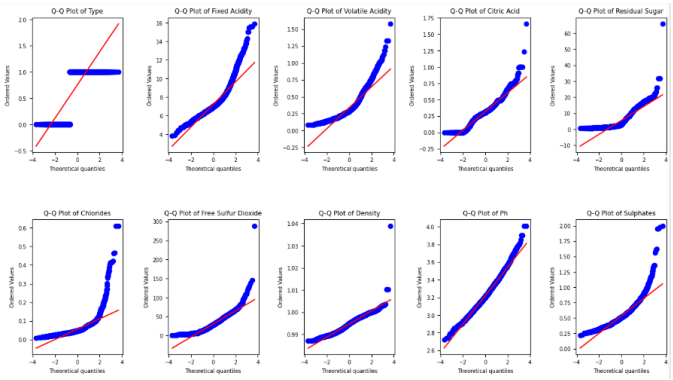


Fig. 4. Q-Q plot assessing normality of predictor variables; deviations from the line suggest non-Gaussian distributions.

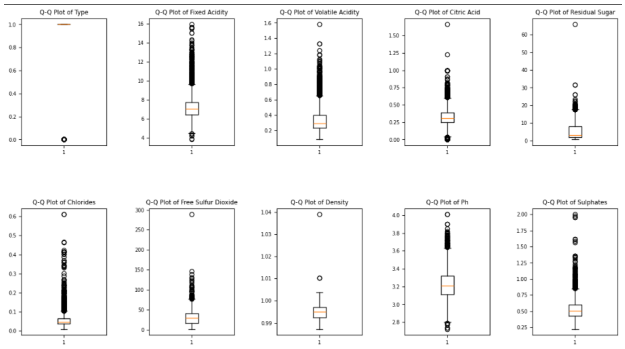


Fig. 5. Boxplot of predictor variables showing the spread, central tendency, and presence of outliers across key physicochemical features.

b) Data Splitting and Scaling: The dataset was randomly split into training and testing subsets using an 80%/20% ratio. The training set was used to fit the model, while the test set provided an unbiased evaluation of its generalization performance.

Feature scaling was not applied, as Random Forests are tree-based models and inherently invariant to the scale of input variables. Unlike distance-based algorithms such as k-Nearest Neighbors or Support Vector Machines, Random Forests can effectively process raw, unscaled numerical features. As such, model training proceeded without normalization or standardization.

c) Model Training: The Random Forest classifier was initially trained using 80% of the dataset, with 100 estimators and a fixed random state for reproducibility. To address class imbalance, we applied the Synthetic Minority Oversampling Technique (SMOTE) after merging quality level 9 with class 8 due to its extremely low representation. This resampling improved the representation of minority classes in the training set. To further enhance generalization, we used five-fold cross-validation and employed `RandomizedSearchCV` to optimize hyperparameters, including the number of estimators, maximum depth, and minimum samples per split. This tuning helped reduce overfitting and produced a more balanced model capable of better handling rare wine quality classes.

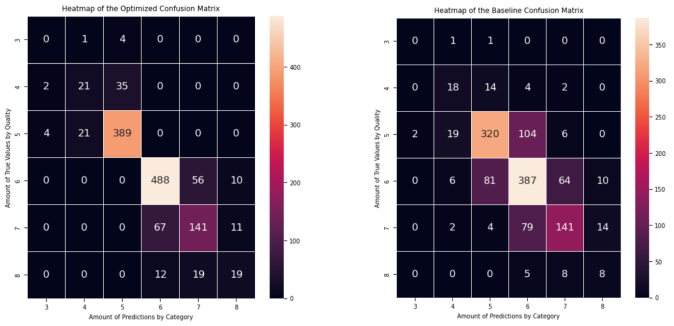
IV. RESULTS AND DISCUSSION

A. Baseline Model Performance and Error Analysis

The baseline Random Forest classifier achieved an overall test accuracy of 83.69%. While this suggests strong predictive performance, accuracy alone can be misleading due to the dataset's class imbalance. As shown in the classification report, the model performs well on majority classes such as qualities 5 and 6, with F1-scores of 0.95 and 0.87, respectively. Despite SMOTE and tuning, the model failed to classify any wines rated as quality 3. The marginal F1-score gain (0.52 to 0.55) highlights the challenge of modeling underrepresented classes. The baseline model was evaluated on a test set of 1,300 samples with distribution reflecting the dataset's inherent imbalance: quality 3 (2 samples), quality 4 (46 samples), quality 5 (420 samples), quality 6 (579 samples), quality 7 (221 samples), and quality 8 (32 samples).

The confusion matrix (Figure 6) reveals that most misclassifications occur between adjacent quality scores, such as between classes 5 and 6. This reflects the subtle nature of quality differentiation and the ordinal structure of the labels. The model makes few to no correct predictions for rare classes like 3 and 8, highlighting its limited ability to generalize to underrepresented categories. The pronounced diagonal in the matrix confirms that performance is biased toward high-frequency classes.

a) Evaluation Metrics and Class Imbalance Considerations: The imbalanced distribution of wine quality classes posed a significant modeling challenge. A model can achieve deceptively high overall accuracy by disproportionately favoring the majority classes (e.g., qualities 5 and 6), while



(a) Optimized model confusion matrix. Improved recall for minority classes.

(b) Baseline model confusion matrix. Misclassifications mostly between adjacent classes.

Fig. 6. Comparison of confusion matrices for baseline and optimized Random Forest models.

performing poorly on underrepresented categories such as qualities 3 and 9. As a result, accuracy alone was deemed insufficient for evaluating model performance. To address this, we report class-wise precision, recall, and F1-score, as shown in Table I. Precision measures the proportion of correct positive predictions, while recall quantifies the model's ability to identify all instances of a given class. The F1-score, as the harmonic mean of precision and recall, provides a more balanced view of classification performance—particularly important in imbalanced settings.

The macro-average F1-score (0.52) highlights the model's uneven performance across classes, as it gives equal weight to all categories. In contrast, the weighted-average F1-score (0.82) reflects the dominance of high-frequency classes. Together, these metrics offer a comprehensive view of the model's strengths and limitations in imbalanced classification tasks. Based on the confusion matrix, the baseline Random Forest model correctly classified 0 instances of quality 3, 7 of quality 4, 419 of quality 5, 454 of quality 6, 109 of quality 7, and 8 of quality 8. The classification report confirms that the model failed to identify class 3 entirely and exhibited limited success with classes 4, 7, and 8. In contrast, it performed strongly on the more frequent classes 5 and 6. Most misclassifications occurred between adjacent quality levels, reflecting the ordinal structure of the labels and the subtle differences between neighboring classes. These results indicate that while the model is effective for predicting common wine qualities, it struggles to generalize to underrepresented categories.

B. Optimized Model Performance

To address the limitations of the baseline model, we applied the Synthetic Minority Oversampling Technique (SMOTE) to balance the training data. Class 9, which had only four instances, was merged with class 8 to ensure sufficient representation during resampling. Median imputation was again used to handle missing values, consistent with the preprocessing pipeline. To further improve performance, we optimized hyperparameters using `RandomizedSearchCV` with five-fold cross-validation. The search space included parameters

such as the number of estimators, maximum tree depth, minimum samples per split, and minimum samples per leaf. This process mitigated overfitting and yielded a more robust model configuration for imbalanced multiclass classification.

C. Accuracy

After applying SMOTE and hyperparameter tuning, the optimized model achieved an overall test accuracy of 81.3%. While this is slightly lower than the baseline (83.7%), the model demonstrates improved balance in class-wise performance. In particular, classification accuracy for underrepresented classes such as 4 and 8 improved substantially. The confusion matrix (Figure 6) shows a broader diagonal spread across classes compared to the baseline, indicating a more even distribution of correct predictions. Specifically, the optimized model correctly classified 0 instances of quality 3, 21 of quality 4, 389 of quality 5, 488 of quality 6, 141 of quality 7, and 19 of quality 8. Compared to the baseline, this reflects notable gains in classes 4, 6, and 8, while performance on class 7 remained stable.

As shown in Table I, classes 5 and 6 continued to achieve strong F1-scores of 0.92 and 0.87, respectively, confirming the model's reliability in predicting frequent categories. Improvements are also evident in minority classes: class 4 increased its F1-score from 0.29 to 0.42, and class 8 from 0.40 to 0.42. Although performance on class 3 remains poor, the model overall exhibits better generalization and reduced bias toward dominant classes. The macro-average F1-score increased from 0.52 to 0.55, while the weighted average remained stable at 0.82, indicating a more equitable distribution of predictive performance. The optimized model evaluation encompassed 1,300 test samples distributed across quality levels 3-8, with quality 6 (567 samples) and quality 5 (428 samples) representing the majority classes, while extreme ratings remained underrepresented.

V. Model Comparison and Features

A. Model Comparison

Quality	Baseline			Optimized		
	Precision	Recall	F1	Precision	Recall	F1
3	0.00	0.00	0.00	0.00	0.00	0.00
4	0.80	0.17	0.29	0.36	0.49	0.42
5	0.91	1.00	0.95	0.94	0.91	0.92
6	0.82	0.95	0.88	0.88	0.86	0.87
7	0.73	0.50	0.60	0.64	0.65	0.65
8	1.00	0.25	0.40	0.38	0.47	0.42
Macro Avg	0.71	0.48	0.52	0.53	0.56	0.55
Weighted Avg	0.84	0.84	0.82	0.82	0.81	0.82

TABLE I
SIDE-BY-SIDE COMPARISON OF CLASSIFICATION METRICS (PRECISION, RECALL, F1-SCORE) ACROSS WINE QUALITY LEVELS FOR BASELINE AND OPTIMIZED MODELS.

Table I compares class-wise performance between the baseline and optimized models. The optimized model exhibits improved recall and F1-score for underrepresented classes, particularly quality levels 4 and 8, whose F1-scores rose from 0.29 to 0.42 and from 0.40 to 0.42, respectively. Although precision slightly declined for some classes, the improvements in recall indicate better identification of minority classes. The macro-average F1-score increased from 0.52 to 0.55, reflecting more balanced performance across all quality levels. The weighted-average F1-score remained stable at 0.82, indicating that gains in minority class performance were not achieved at the expense of majority class accuracy. Overall, the optimized model demonstrates enhanced generalization, offering a more inclusive classification across the wine quality spectrum.

B. Feature Importance Analysis

Permutation importance analysis was used to interpret the contribution of individual features to model predictions. As shown in Table II, alcohol emerged as the most influential predictor, followed by volatile acidity, free sulfur dioxide, density, and chlorides. These findings are consistent with enological literature. According to Waterhouse [9] alcohol content is a key indicator of grape ripeness and fermentation outcomes, contributing to structure and mouthfeel. Volatile acidity affects aroma and taste, where lower levels are generally preferred for a cleaner profile. Free sulfur dioxide serves as a preservative, maintaining freshness and preventing oxidation. Density reflects residual sugar and alcohol content, indirectly capturing fermentation efficiency. Chlorides, indicative of sodium content, influence taste balance and chemical stability.

Other features such as pH were found to be less impactful in this model, suggesting a secondary role in driving predicted quality scores.

Rank	Feature	Importance Score
1	Alcohol	0.14
2	Volatile Acidity	0.08
3	Free Sulfur Dioxide	0.04
4	Density	0.03
5	Chlorides	0.02

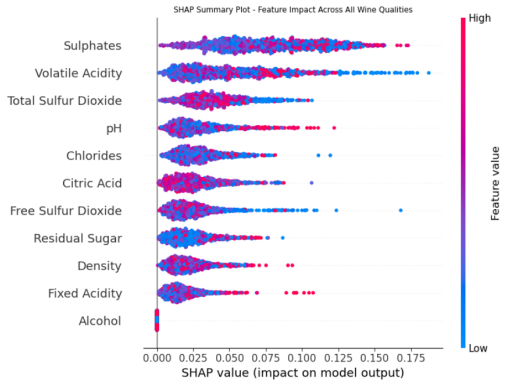
TABLE II
TOP 5 FEATURES RANKED BY PERMUTATION IMPORTANCE.

C. Model Interpretability

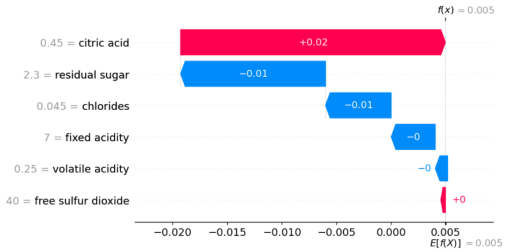
Beyond permutation importance, we applied SHAP (SHapley Additive exPlanations) to derive global and local insights into our Random Forest model. SHAP values estimate each feature's marginal contribution to individual predictions relative to a baseline. The summary plot in Figure 7 highlights volatile acidity and free sulfur dioxide as globally dominant predictors—consistent with permutation importance. However, alcohol, ranked highest in permutation, exhibits limited global SHAP impact. This contrast illustrates the need to examine both model-agnostic and model-consistent interpretability. To reconcile these perspectives, we present a local SHAP waterfall plot for a high-quality prediction. It shows alcohol making a strong positive contribution when volatile acidity is low, suggesting its influence is conditional rather than uniform. This observation aligns with oenological theory on wine balance and complexity.

VI. Conclusion and Future Work

This study illustrates the potential of machine learning to enhance traditional wine quality evaluation through an



(a) Global importance (mean impact)



(b) Local explanation (single prediction)

Fig. 7. SHAP-based interpretability of Random Forest predictions. **(a)** Global feature importance across all predictions. Volatile acidity and free sulfur dioxide dominate, while alcohol shows limited average contribution. **(b)** Local SHAP explanation for a specific instance, illustrating how individual feature values influence the prediction.

interpretable, data-driven, and reproducible framework. Using a Random Forest classifier trained on physicochemical properties from the UCI Wine Quality dataset, we achieved a baseline accuracy of 83.7%. The model exhibited strong performance on majority classes but showed limitations when predicting underrepresented categories—highlighting a need for greater robustness in imbalanced classification tasks. In addition to predictive performance, we conducted a detailed feature importance analysis using permutation techniques, identifying alcohol, volatile acidity, and free sulfur dioxide as the most impactful predictors—results that are consistent with oenological domain knowledge. The modeling pipeline emphasized clarity in visual communication and reproducibility, providing a foundation that can be extended by researchers and adapted by practitioners in the wine industry.The complete pipeline was implemented in Python using scikit-learn and is available for reproducibility at https://github.com/cejones8/Wine-Quality-Prediction.git.

Future work can proceed in several directions. Incorporating external variables such as vineyard location, grape variety, and soil characteristics could enhance predictive accuracy. Advanced optimization techniques including grid search and Bayesian methods may further improve model performance. Given the ordinal structure of wine ratings, future work may explore ordinal classification or regression models to better capture class relationships. Additionally, reframing the classification problem as a regression task could better reflect the ordinal and continuous nature of wine quality, potentially improving prediction resolution and usability. Interactive deployment through web and mobile applications would facilitate real-time quality prediction, while API integration with laboratory systems could provide immediate SHAP-based explanations for industrial applications.

A. Broader Impacts

This framework enables rapid, objective wine quality evaluation, reducing reliance on subjective tasting panels while improving consistency and confidence in quality labeling. From a consumer perspective, deploying the model as interactive tools democratizes access to wine quality insights, enhancing transparency in wine selection and empowering consumers with objective information for purchasing decisions. For industry deployment, the framework could integrate with existing laboratory systems via API, providing real-time quality predictions with SHAP explanations through web or mobile interfaces. Economic benefits include reduced quality control costs and improved consistency across production batches. The modular Python pipeline enables seamless integration into winery workflows with minimal infrastructure requirements. The methodology extends to other agricultural products (coffee, olive oil, dairy), promoting scalable, data-driven quality assurance across food industries.

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